

ATTACK TYPE	SEVERITY	CONFIDENCE	SUBTYPE
SQLi	CRITICAL	90.4%	UNION-based SQLi

[WARNING] SQLi detected with subtype: UNION-based SQLi

Payload

```
' UNION SELECT username,password FROM users --
```

Request Metadata

Field	Value
ip	127.0.0.1
user_agent	Mozilla/5.0 (Windows NT; Windows NT 10.0; en-US) WindowsPowerShell/5.1.26100.8894
path	/api/xai/report
method	POST

Attack-Specific Explanation

- The payload contains UNION SELECT style syntax, commonly used to merge attacker queries with application queries.
- This often indicates an attempt to extract additional database rows or columns.
- Strong matched indicators: union_select, select_from
- Weak matched indicators: select_keyword, union_keyword, comment_end

Top TF-IDF Terms (Why It Was Flagged)

Token	TF-IDF Score	Visual
user	0.1987	#####
use	0.1939	##### -
us	0.1898	##### -
serna	0.1660	##### - - -
usern	0.1660	##### - - -
sern	0.1660	##### - - -
rname	0.1642	##### - - -
rnam	0.1642	##### - - -
ernam	0.1642	##### - - -
word	0.1582	##### - - - -

Potential Impact & Recommended Protections

Potential Impact	Recommended Protections
- Authentication bypass, unauthorized data access, or database fingerprinting.	- Use parameterized queries or prepared statements everywhere.
- Potential data extraction, modification, or destruction depending on permissions.	- Never concatenate untrusted input into SQL queries.
	- Apply least-privilege database accounts and suppress verbose DB errors.

Attack Preconditions

1	User-controlled input reaches an SQL context without proper parameter binding.
2	The backend query construction allows attacker input to alter query structure.

Decision Debug / AI Gating

Key	Value
raw_pred	1

Key	Value
proba	{ "0": 0.096, "1": 0.904, "2": 0.0, "3": 0.0 }
pattern_hits	{ "sqli": { "strong": ["union_select", "select_from"], "weak": ["select_keyword", "union_keyword", "comment_end"] }, "xss": { "strong": [], "weak": [] }, "lfi": { "strong": [], "weak": [] } }
decision_reason	multi_allowed_choose_highest
thresholds	{ "sqli": { "no_pattern_min": 0.85, "weak_min": 0.8, "strong_min": 0.6 }, "xss": { "no_pattern_min": 0.85, "weak_min": 0.8, "strong_min": 0.6 }, "lfi": { "no_pattern_min": 0.85, "weak_min": 0.8, "strong_min": 0.6 } }

Report Note

- This report was automatically generated by the HWACS Explainable AI Security Platform using captured request data, model prediction output, pattern-based analysis, subtype detection, and explainability signals. The result should be reviewed by a qualified security professional before taking operational action.